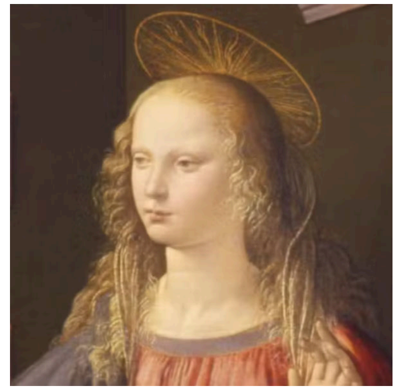
Leonardo's study of shadow led to a breakthrough in his early portrait of *Ginevra de'Benci*. In the *Annunciation*, despite refinement, the Virgin remains somewhat stiff. With *Ginevra*, Leonardo pursued another goal described in the *Trattato*



> A picture, or representation of the human figure, ought to be done in such a way that the spectator may easily recognize the motions of the mind by the attitudes of the body.

Most portraits of young women in Florence were painted in strict profile – a format that limits psychological expression. For Ginevra, Leonardo chose a three-quarter, bringing beholder and sitter in dialogue. This orientation allowed him to sculpt the shadow of every facial feature. Through gradations of light and shade, and achieved a convincing softness in her young face and introduced a new level of psychological presence to portraiture. An early biographer wrote that the portrait was "painted with such perfection that it was none other than she."

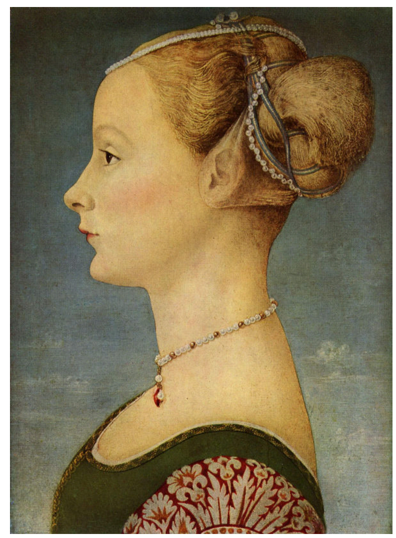


Figure 61: *Portrait of a Young Lady* by Piero del Pollaiolo, 1470. Link to Museo Poldi Pezzoli

Figure 62: *Ginevra de'Benci* by Leonardo da Vinci, 1474-1478 Link to NGA

THE MANNERIST style followed upon the High Renaissance and reacted against its idealized harmony. Mannerists sought realism, but enhanced for dramatic effect. Giovanni Battista Moroni (1520 - 1578), in his *Portrait of a Gentleman*, is an analytical depiction of shadow. He compares a variety of cast and modeling shadows across convex and concave surfaces (Fig. 63). Whereas Leonardo softened transitions, Moroni sharpens boundaries between light and dark to emphasize physical realism.

Figure 63: *Portrait of a Gentleman*, Giovanni Battista Moroni, 1556. National Gallery

MICHELANGELO CARAVAGGIO (1571-1610) pushed shadow towards dramatic extremes. His followers, known as *tenebrists*, embraced stark contrasts between light and darkness. Caravaggio revolutionized religious painting by using shadow not merely to model form, but to insist on reality.

In *The Incredulity of Saint Thomas*, Caravaggio renders the biblical scene with brutal realism. Light cuts across wrinkled brows and weathered skin.

> Then saith he to Thomas, Reach hither thy finger, and behold my hands; and reach hither thy hand, and thrust it into my side: and be not faithless, but believing.
> – John 20:27 (KJV)

Thomas presses his finger into Christ's wound. Harsh, theatrical lighting intensifies the physicality of the encounter. Sacred figures are portrayed as ordinary men – embodied and tangible.

Figure 64: *The Unbelieving Thomas*, Caravaggio, 1601. Sarassouci Museum

ITALIAN DRAMATIC LIGHTING influenced Northern Europe. Rembrandt and his circle embraced shadow as expressive force. In works from his school, raking light streams through windows, casting elongated shadows that structure entire compositions. Paintings become meditations on the interplay between illumination and obscurity.

Figure 65: *Man Seated Reading at a Table in a Lofty Room.* by Follower of Rembrandt or his school, 1628/30. Link to painting at the National Gallery.

SILHOUETTE belongs to the origins of art, but experienced remarkable popularity in eighteenth- and nineteenth-century America. Skilled artists could cut convincing profiles with scissors alone. Technological devices soon accelerated production, making silhouettes affordable and widely accessible.

One such invention, the physiognotrace, allowed individuals to trace profile outlines mechanically. Silhouette became a democratizing art form—economical, reproducible, and ennobling. Anyone, from ordinary citizen to President, could possess a likeness in profile.

The term "silhouette" derives from Étienne de Silhouette (1709-1761), a French finance minister reputed for frugality. The name attached itself to this economical form of portraiture.

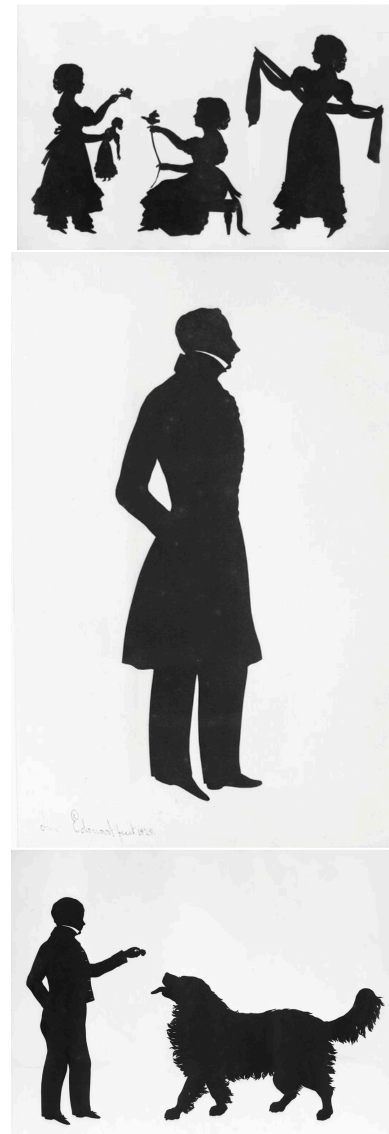


Figure 66: Augustin Amant Constant Fidèle Edouart, French, *Album of Silhouettes*. Harvard Art Museums.

CHEMICAL PHOTOGRAPHY would end the brief but surging popularity of the American portrait silhouette. But the silhouette retains a timeless quality, and a perspective on other art forms by comparison. When an image is flattened and turned into black and white, all sense of depth and color are lost. The remaining silhouette keeps a sense of realism as pure shadow.

Figure 67: A Sure and Convenient Machine for Drawing Silhouettes by Thomas Holloway ca. 1788.

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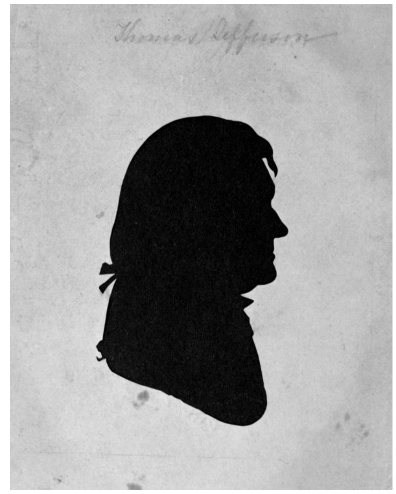


Figure 68: *Jefferson Portrait* by Raphaelle Peale 1798. Monticello Museum.

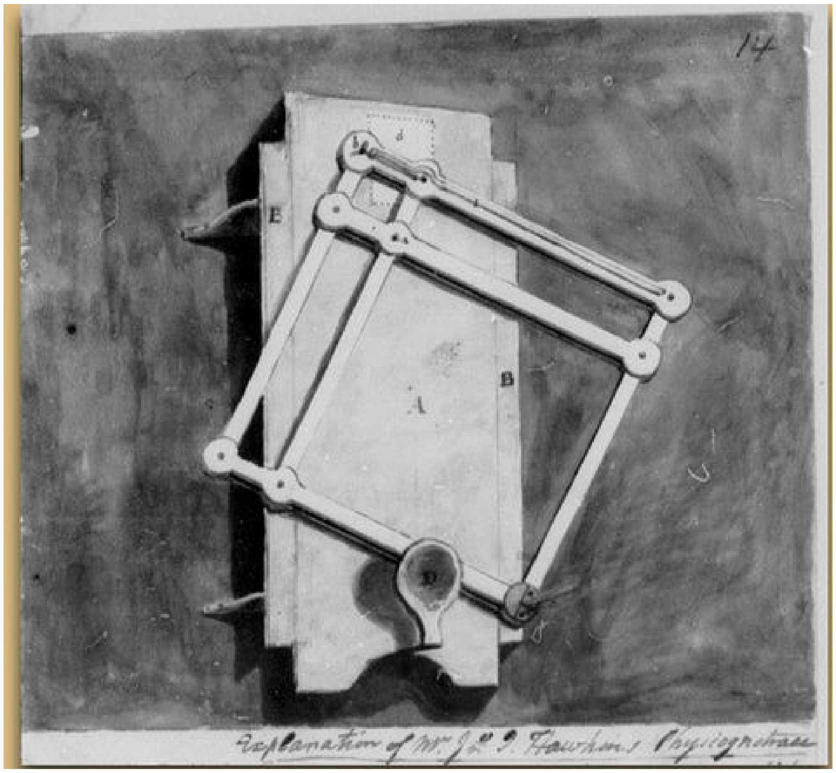
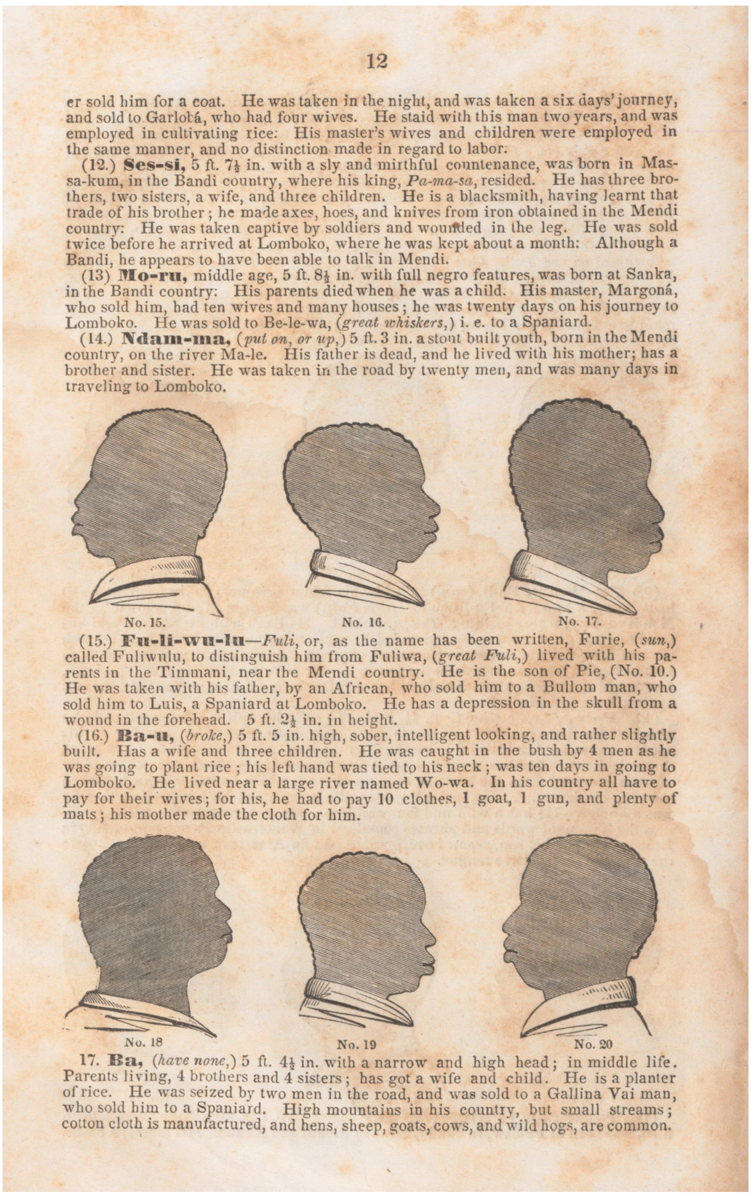


Figure 69: Drawing of John Hawkin's physiognotrace device.

Physiognomic theories – racism couched in the scientific language of 'phrenology' – held that the profile and shape of the skull were indicators of intelligence and character. Thus, silhouette could be connected with troubling and debunked scientific theory.

La Amistad was the slave ship that saw revolt in 1839. It was captured in New York and the case of the slaves was decided in *United States vs. The Amistad* in favor of freedom. However, in John Warner Barber's *History of the Amistad Captives*, the identities of the 36 captives is shown only in grotesquely reductive silhouette, along with words evoking phrenology.



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er sold him for a coat. He was taken in the night, and was taken a six days' journey, and sold to Garlobá, who had four wives. He staid with this man two years, and was employed in cultivating rice: His master's wives and children were employed in the same manner, and no distinction made in regard to labor:

(12.) **Ses-si,** 5 ft. 7½ in. with a sly and mirthful countenance, was born in Mas-sa-kum, in the Bandi country, where his king, *Pa-ma-sa,* resided. He has three brothers, two sisters, a wife, and three children. He is a blacksmith, having learnt that trade of his brother; he made axes, hoes, and knives from iron obtained in the Mendi country: He was taken captive by soldiers and wounded in the leg. He was sold twice before he arrived at Lomboko, where he was kept about a month: Although a Bandi, he appears to have been able to talk in Mendi.

(13) **Mo-ru,** middle age, 5 ft. 8½ in. with full negro features, was born at Sanka, in the Bandi country: His parents died when he was a child. His master, Margoná, who sold him, had ten wives and many houses; he was twenty days on his journey to Lomboko. He was sold to Be-le-wa, (*great whiskers,*) i. e. to a Spaniard.

(14.) **Ndam-ma,** (*put on, or up,*) 5 ft. 3 in. a stout built youth, born in the Mendi country, on the river Ma-le. His father is dead, and he lived with his mother; has a brother and sister. He was taken in the road by twenty men, and was many days in traveling to Lomboko.

No. 15. No. 16. No. 17.

(15.) **Fu-li-wu-lu**—*Fuli*, or, as the name has been written, Furie, (*sun,*) called Fuliwulu, to distinguish him from Fuliwa, (*great Fuli,*) lived with his parents in the Timmani, near the Mendi country. He is the son of Pie, (No. 10.) He was taken with his father, by an African, who sold him to a Bullom man, who sold him to Luis, a Spaniard at Lomboko. He has a depression in the skull from a wound in the forehead. 5 ft. 2½ in. in height.

(16.) **Ba-u,** (*broke,*) 5 ft. 5 in. high, sober, intelligent looking, and rather slightly built. Has a wife and three children. He was caught in the bush by 4 men as he was going to plant rice; his left hand was tied to his neck; was ten days in going to Lomboko. He lived near a large river named Wo-wa. In his country all have to pay for their wives; for his, he had to pay 10 clothes, 1 goat, 1 gun, and plenty of mats; his mother made the cloth for him.

No. 18 No. 19 No. 20

17. **Ba,** (*have none,*) 5 ft. 4½ in. with a narrow and high head; in middle life. Parents living, 4 brothers and 4 sisters; has got a wife and child. He is a planter of rice. He was seized by two men in the road, and was sold to a Gallina Vai man, who sold him to a Spaniard. High mountains in his country, but small streams; cotton cloth is manufactured, and hens, sheep, goats, cows, and wild hogs, are common.

Figure 70: *A History of the Amistad Captives: Being a Circumstantial Account of the Capture of the Spanish Schooner Amistad, by the Africans on Board; Their Voyage, and Capture near Long Island, New York; with Biographical Sketches of Each of the Surviving Africans.* by John Warner Barber, 1840. Metropolitan Museum

A SILHOUETTE PROFILE can capture identity with precision, but it is also inherently anonymizing. When a silhouette is made from any direction but side profile, identity dissolves. The blank "profile icon" used in social media is a contemporary skeuomorph of the silhouette – an abstract head awaiting identity.

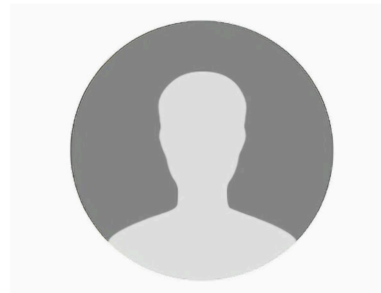

Figure 71: Anonymous

THE AESTHETIC OF ANONYMITY, using a silhouette not painted in profile, was used by artists. Gerard ter Borch (1617-1681), a contemporary of Vermeer, paints an unknown and unknowable soldier in his grim *Man on Horseback*.

Figure 72: *Man on Horseback* by Gerard ter Borch, 1634. MFA

MODERN ARTISTS have reinterpreted the silhouette as a symbolic form. Kara Walker manipulates the history of the silhouette in her social commentaries (Fig. 74). In contrast to polite portraiture silhouette from the 19th century, Walker evokes the exploitation and dark fantasy of 19th century life.

Figure 73: *African/American* by Kara Walker, 1998. Whitney Museum

Figure 74: *Auntie Walkers Wall Sampler for Savages* by Kara Walker. Kara Walker, a contemporary artist, a mural-scale silhouettes that enacts the cruelty of antebellum plantation life.

KUMI YAMASHITA plays with the dimensionality of the silhouette by shaping paper to create silhouettes as shadows when properly illuminated with raking light(Fig. 75). Yamashita's works are anamophoric shadows. The shadows make you think about how they were made.

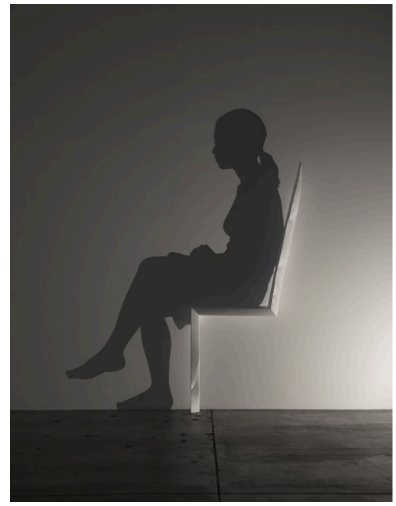

Figure 75: *Chair*, by Kumi Yamashita, 2014. Carved wood, single light source, shadow.

Figure 76: *Fragments*, by Kumi Yamashita, 2009.

FRANCESCA BEWER, the Director of the M-Lab at the Harvard Art Museums, made a surprisingly complex shadow in her photographic still life of lemons.

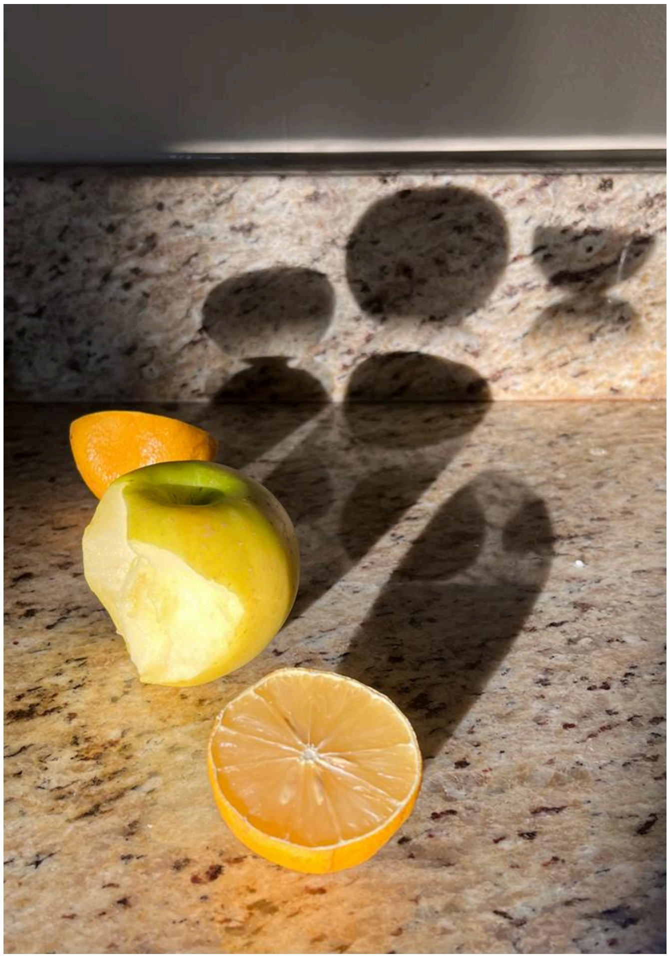

Figure 77: *Still Life with Oranges* by Francesca Bewer, 2024.

Bibliography

- E. H. (Ernst Hans) Gombrich. *Shadows : the depiction of cast shadows in western art*. Yale University Press, New Haven, new edition. edition, 2014. ISBN 9780300210040
- Asma Naeem. *Black Out: Silhouettes Then and Now*. Princeton University Press, Princeton, New Jersey, 2018. ISBN 069118058X
- Francesca Fiorani. *The shadow drawing : how science taught Leonardo how to paint*. Farrar, Straus and Giroux, New York, first edition. edition, 2020. ISBN 9780374261962

Updated: February 19, 2026

Seeing Perspective

Thursday, 19 February 2026, 12:45 - 2:15 PM EST.
Harvard Art Museums 0600

We see the three-dimensional world through visual information that has been flattened onto the retina. The brain unconsciously expands this two-dimensional input into a sense of depth using cues including relative size, shadow and illumination, and the occlusion of distant objects by nearer ones. When we look at two-dimensional pictures, our brains naturally identify these depth cues, allowing us to infer the three-dimensional spaces they depict.

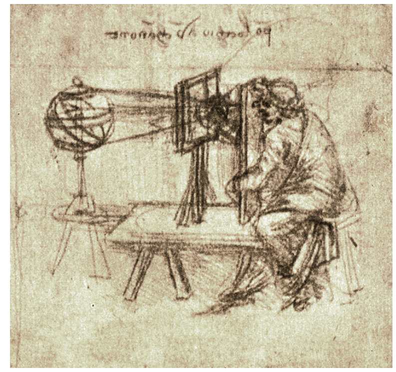

Figure 78: Leonardo's *Draughtsman using a Transparent Plane to Draw an Armillary Sphere*

But to make a picture, the artist must consciously translate three-dimensional information into a two-dimensional image. To succeed, the artist often must work against their own unconscious visual processing.

The word *perspective* derives from the Latin *perspicere*, meaning 'to see through' or 'to perceive clearly'. Its meanings changed over time, but it came to describe a geometrically consistent method for representing three-dimensional scenes on a flat surface, first demonstrated in Florence by Filippo Brunelleschi (1377–1446). This method, now called *linear perspective*, is illustrated in Leonardo da Vinci's drawing of a draughtsman projecting a three-dimensional sphere onto a transparent plane.

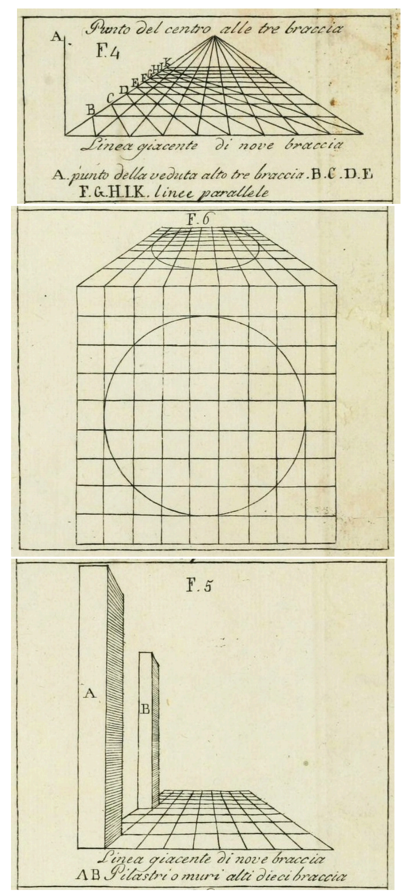

Figure 79: Figures from the 1804 edition of *De Pictura* showing the vanishing point; how a circle is projected as an ellipse; showing the diminishing size of pillars viewed at a distance.

In simplified terms, Brunelleschi discovered what happens when a rectilinear three-dimensional space is projected onto a two-dimensional plane from a fixed eye-point. When every point in space is projected onto the picture plane along lines that pass through this single eye-point, parallel lines orthogonal to the viewing plane converge to a single vanishing point. Other sets of parallel lines likewise converge, each to its own vanishing point.

If a set of parallel lines forms a 45-degree angle with the picture plane, then the lateral vanishing point lies at a horizontal distance from the central vanishing point equal to the distance between the eye-point and the picture plane. If both central and lateral vanishing points can be located in a painting, the intended viewing position can, in principle, be reconstructed.

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Leon Battista Alberti (1404–1472) codified Brunelleschi's discovery in *De Pictura* (1435). Once formulated in theoretical terms, linear perspective spread rapidly through Western art. Its geometric clarity established principles that artists continued to consult for centuries. Vermeer, for example, owned a copy of *The Book of Perspective* by Hans Vredeman de Vries.

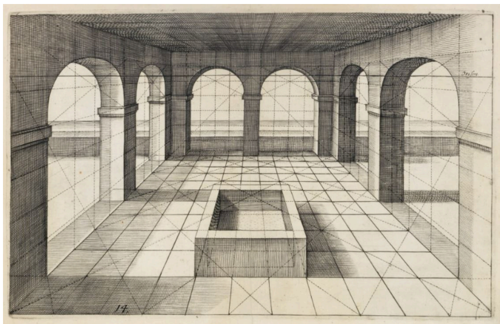


Figure 80: An engraving from *The Book of Perspective* by Hans Vredeman de Vries. De Vries wrote and illustrated a guidebook on perspective that artists of the day, including Vermeer, owned and consulted.

Despite the sophistication of Greek and Roman art and science, there is no evidence that antiquity used Brunelleschian single-point perspective as later codified. Greek geometrical theories of vision – especially Euclid's *Optics* – modeled sight as rays extending from the eye. In his Eighth Theorem, Euclid writes:

> Of equal magnitudes situated at unequal distances, the nearer appears larger, and the more distant appears smaller.

Euclid explains apparent size in terms of the visual angle subtended at the eye. While he modeled rays geometrically in space, he did not conceptualize an image-forming projection plane between object and observer. Brunelleschi's breakthrough required shifting attention from angular magnitude at the eye to projection onto a plane within space.

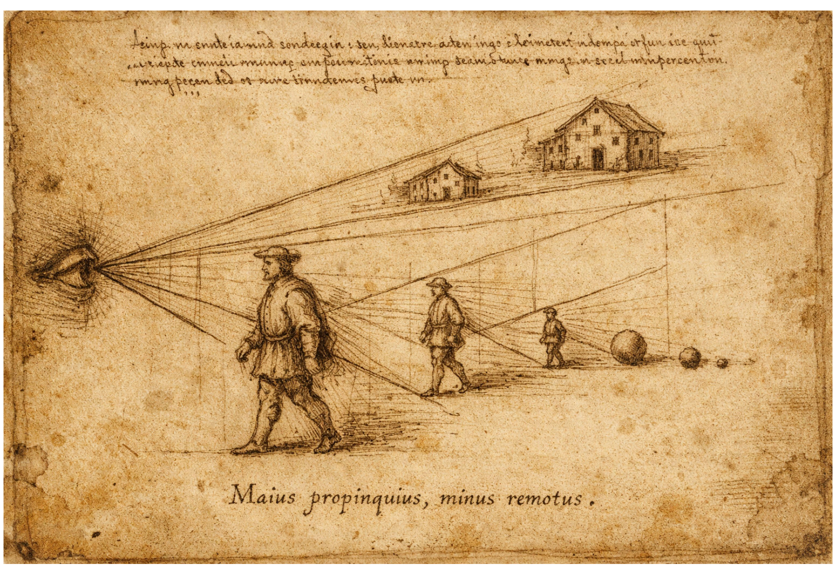


Figure 81: Diagram of the principle that equal objects seen at unequal distances subtend unequal visual angles. Rays issuing from the eye form visual cones to each object; the nearer figure intercepts a wider angle and therefore appears larger, while the more distant figures intercept progressively narrower angles and appear smaller. The drawing visualizes Euclid's claim that apparent magnitude depends not on intrinsic size but on angular magnitude at the eye.

Another interpretive obstacle may have been cosmological. In antiquity and the Middle Ages, the universe was commonly conceived as a finite cosmic sphere. Erwin Panofsky suggests that this differs fundamentally from the later assumption of infinite, homogeneous space implicit in linear perspective. Renaissance depictions of the Ptolemaic cosmos – such as Giovanni di Paolo's fifteenth-century rendering – visualize a finite celestial system structured by concentric spheres.

Figure 82: 16th-century representation of Ptolemy's geocentric model in Peter Apian's *Cosmographia* , 1524

The conceptual shift toward imagining space as continuous and homogeneous may have prepared the ground for perspectival projection. Antique structural assumptions of space had to be reconfigured. Italian Gothic painting, such as Cimabue's *Maestá* (c. 1280), does not construct a unified geometric space. Figures vary in scale according to symbolic hierarchy rather than strict optical proportion. Gold grounds flatten the field, and recession is not organized around a single vanishing system.

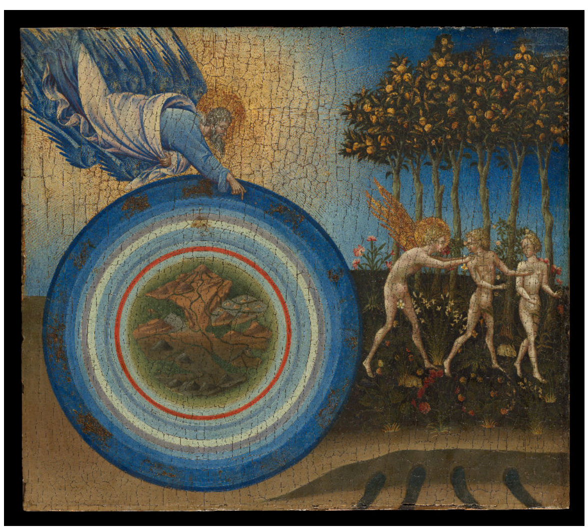
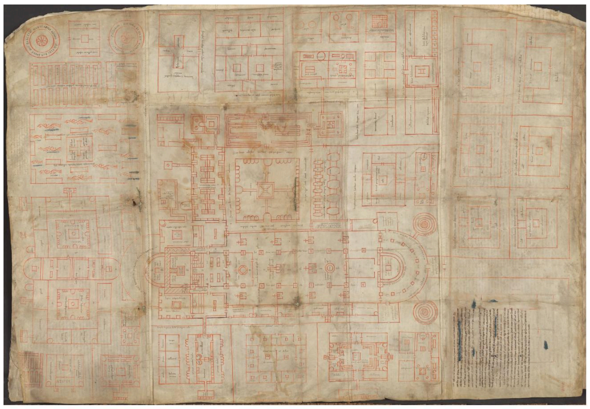


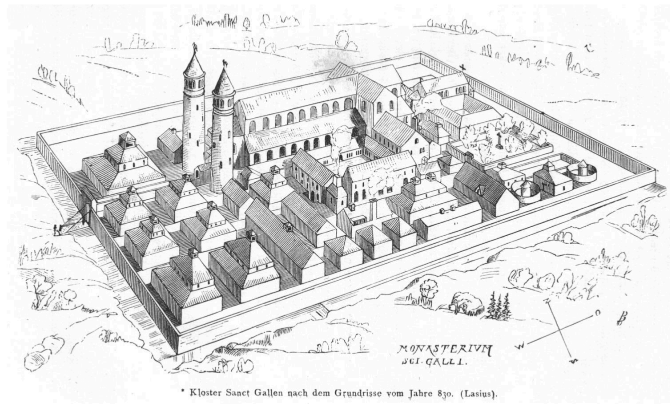
Figure 84: *The Creation of the World and the Expulsion from Paradise* by Giovanni di Paolo (1445) Link to Met

Figure 83: *La Vierge et l'Enfant en majesté entourés de six anges* by Cimabue Link to Louvre

It may not be surprising that Brunelleschi, trained as an architect, first formalized linear perspective. Architects routinely work with orthographic projection—plans in which parallel lines remain parallel and objects are drawn to scale. The Plan of St. Gall (c. 820–830), an ideal Benedictine monastery rendered in bird's-eye orthographic view, represents functional organization rather than optical appearance. Later reconstructions translate this schematic plan into volumetric and perspectival space.



Plan of St. Gall (Codex Sangallensis 1092, recto)



Rudolf Rahn Reconstruction (1876)

Figure 85: Left: Plan of St. Gall (Codex Sangallensis 1092, recto), c. 820–830, an idealized Benedictine monastery rendered in orthographic plan, representing functional order rather than optical appearance. Right: Rudolf Rahn, reconstruction of the monastery according to the ninth-century plan (1876), translating the schematic ground plan into a unified, modern linear perspective view.

Figure 86: Santa Maria del Fiore, viewed from the Michelangelo Hill. Florence from the south bank of the Arno. Brunelleschi's dome dominates the cityscape, rising above Giotto's Campanile and tiled roofs.

Brunelleschi's architectural career culminated in the dome of Santa Maria del Fiore, still the largest masonry dome ever constructed.

Cathedral architecture played a formative role in the development of perspective. Painters in Italy and Northern Europe increasingly staged sacred scenes within contemporary architectural settings. The complex geometries of Gothic vaults, arcades, and pavements posed challenges in rendering spatial coherence.

Jan van Eyck's *Madonna in a Church* (c. 1438–40) approaches a unified spatial illusion without achieving mathematical consistency. Architectural lines appear to converge, but not consistently to a single vanishing point. The space is visually persuasive yet empirically constructed. Mary appears disproportionately large relative to the architecture, suggesting symbolic hierarchy rather than strict perspectival scale.

Rogier van der Weyden, a contemporary of van Eyck, makes similar adjustments in his *Seven Sacraments Altarpiece*, where space appears coherent but is not mathematically unified.

Figure 87: Left: Jan van Eyck, *Madonna in the Church*, c. 1438–1440. Gem aldegalerie, Staatliche Museen zu Berlin. Link to museum Right: Rogier van der Weyden, *The Seven Sacraments Altarpiece*, c. 1445–1450. Royal Museum of Fine Arts Antwerp (KMSKA). Link to museum

Brunelleschi's perspectival experiments are described by Antonio Manetti. In his Baptistery panel, Brunelleschi painted the building and drilled a small hole at the viewing point. The viewer looked through the hole toward a mirror reflecting the painted image. When aligned with the real building, the painted projection coincided with reality from that fixed vantage point. In a sense, these panels functioned like an early form of virtual reality, restricting the viewer to a single immersive viewpoint.

A second panel depicted the Piazza della Signoria, incorporating multiple oblique facades into a unified perspectival framework organized around the fixed eye-point.

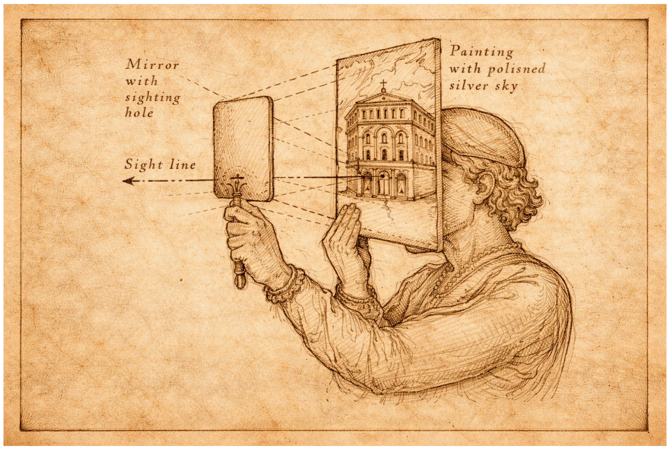


Figure 88: Brunelleschi's Perspective Demonstration (after Antonio Manetti) The viewer looks through a small sighting hole in the back of a painted panel depicting the Florentine Baptistery, while a handheld mirror reflects the image. By aligning the reflected painting with the real building, Brunelleschi demonstrated that a geometrically constructed image could replicate visual reality from a fixed eye-point.

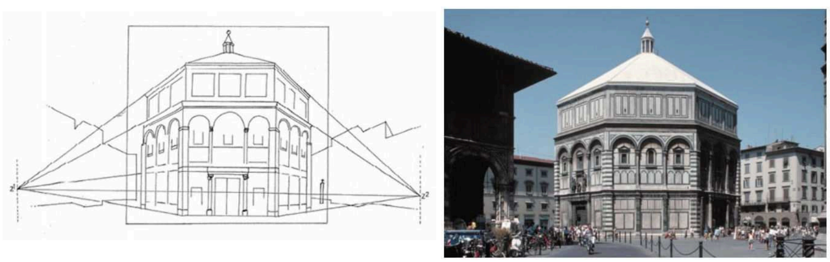


Figure 89: Brunelleschi's first demonstration of linear perspective used the Baptistery of the Florence Cathedral.

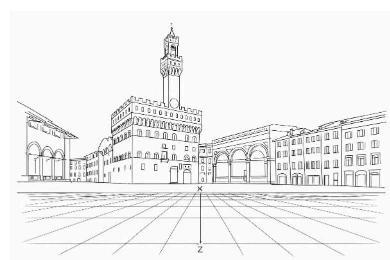


Figure 90: Diagrammatic reconstruction of how Brunelleschi's lost Piazza della Signoria panel might be rendered using the geometric method later codified by Leon Battista Alberti in *De Pictura* .

A second set of panels was made for the Piazza della Signoria in Florence. The Baptistery panel would prove that each set of parallel lines along an object converges to a unified locus. The Piazza panel would be more complex with multiple oblique building facades. Space is not organized around one object, but brings the entire visual scene together into a mathematically coherent framework subjectively ordered by the fixed eye-point of the viewer.

LINEAR PERSPECTIVE implies that painter and viewer should *look* at the image in exactly the same way – one eye in front of the same central point, standing at the same distance from the image – so that both painter and viewer *see* the image in the same way. Any difference between eye locations of the view from the artist would skew what they see. In principle, if both eyes are fixed in the same position, the viewer would have a three dimensional illusion of three-dimensional space. Of course, this wrongly assumes that looking at an even perfect projection onto a two-dimensional plane held at finite distance can match the visual experience of seeing an infinite three-dimensional volume beyond the picture plane.

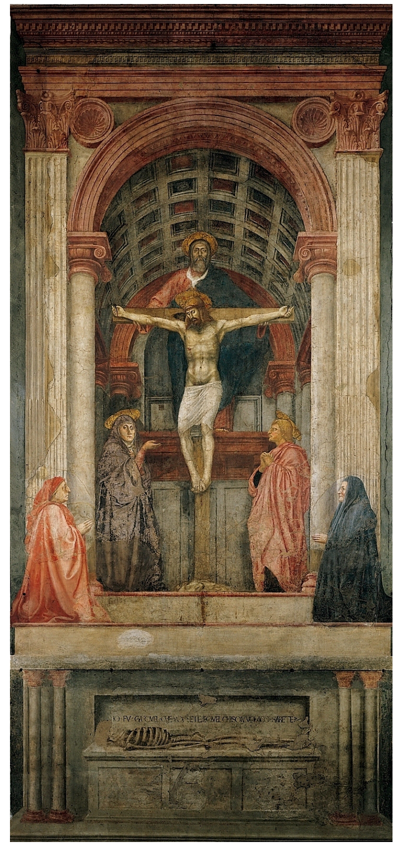
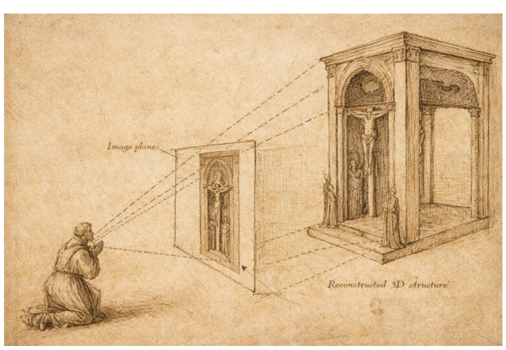
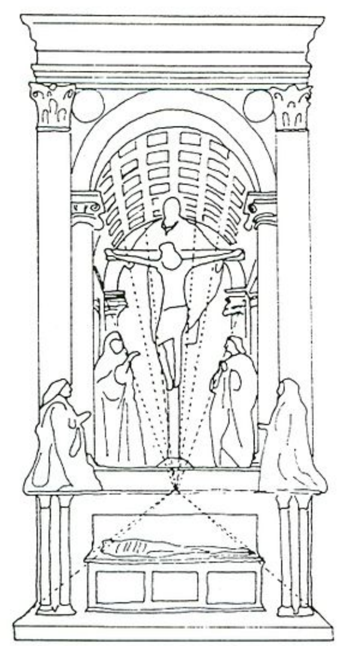


Figure 91: *The Holy Trinity* by Masaccio. Link to artist at Google Arts and Culture.

Masaccio (1401-1428), likely influenced by Brunelleschi, adopted linear perspective in his fresco *The Holy Trinity* — Father, Son, and Holy Ghost symbolized by a white dove — in the Dominican church of Santa Maria Novella in Florence. He constructed an imagined barrel-vaulted chapel beyond the church wall.

In the *Holy Trinity,* Masaccio is not transcribing a real visual scene, like what Brunelleschi did with the Baptistery. Instead, Masaccio first used both mathematical and artistic ingenuity to invent a virtual three-dimensional space in the upper section of the fresco. Masaccio filled this virtual space with a vault with Jesus on the Cross, God the Father standing behind him, Mary and St. John standing on either side, and the donors (who paid for the painting) kneeling just outside the vault. Masaccio imagined the space, and then predicted the projection that would be seen by a kneeling viewer at the altar in front of the huge fresco. This altar – which originally divided the upper section from the lower section – is now gone. But the converging lines in the architecture of the upper section define the line-of-sight that Masaccio intended to give the worshipper an illusion of looking into a vault that extends beyond the wall of the church. In the lower section, below the altar, a skeleton lies in a crypt below the vault floor.





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LINEAR PERSPECTIVE demands geometric rigor, and few artists applied it without compromise. Leonardo da Vinci (1452-1519) and Albrecht Dürer (1471-1528) both explored mechanical devices for transferring spatial information onto flat surfaces. Their notebooks reveal deep theoretical engagement with projection systems.

Figure 92: Albrecht Dürer. 1525. *Artist Drawing a Nude with Perspective Device.*

> "Perspective must... be preferred to all the human discourses and disciplines. In this field of study, the radiant lines are enumerated by means of demonstrations in which are found not only the glories of mathematics but also of physics, each being adorned with the blossoms of the other." -Leonardo da Vinci

Figure 93: Albrecht Dürer. 1525. *Artist Drawing a Vase.*

> "Many years ago I began to understand and learn the art of measurement. I recognized that this knowledge is the foundation of all painting." -Albrecht Dürer

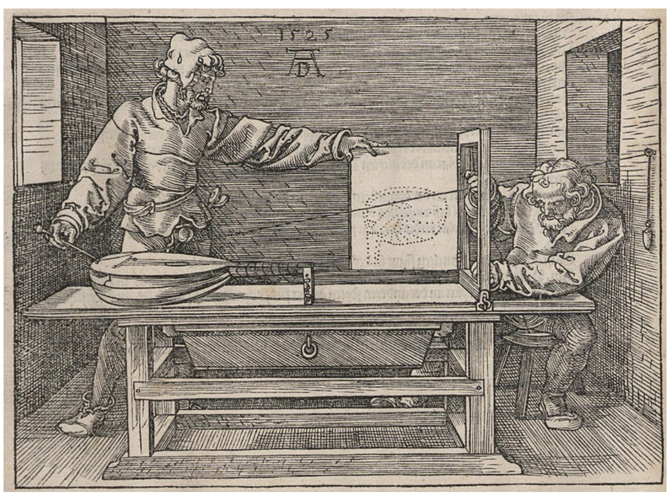

Figure 94: Albrecht Dürer. 1525. *Artist Drawing a Lute.*

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In *The Flagellation*, Piero della Francesca constructed a deep architectural stage populated by two distinct narrative zones. The tiled floor is not a simple checkerboard, but a complex pattern of triangles, squares, rectangles, and parallelograms. Painting this floor, which predated Descartes's Cartesian plane by nearly 200 years, would have required meticulous geometrical planning.

The spatial configuration and shadowing is realistic and coherent throughout *The Flagellation* except for the bright light surrounding Christ and the column to which he is bound. The rest of the scene is illuminated from the left, but Christ is illuminated by a bright light from the right that only he sees. In this way, della Francesca finds a subtle way to emphasize and celebrate a divine light, a private signal from the Father to Son.

Although Piero may have encountered Brunelleschi's ideas in Florence, his own treatise *De Prospectiva Pingendi* demonstrates his independent mathematical mastery.

Figure 95: Piero della Francesca, *Resurrection of Christ*, 1465

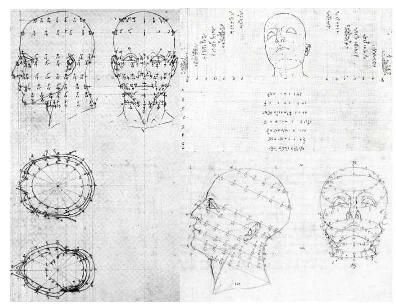

Figure 96: from Piero della Francesca, *De Prospectiva pingendi*, studies that are incorporated in his *Resurrection of Christ*.

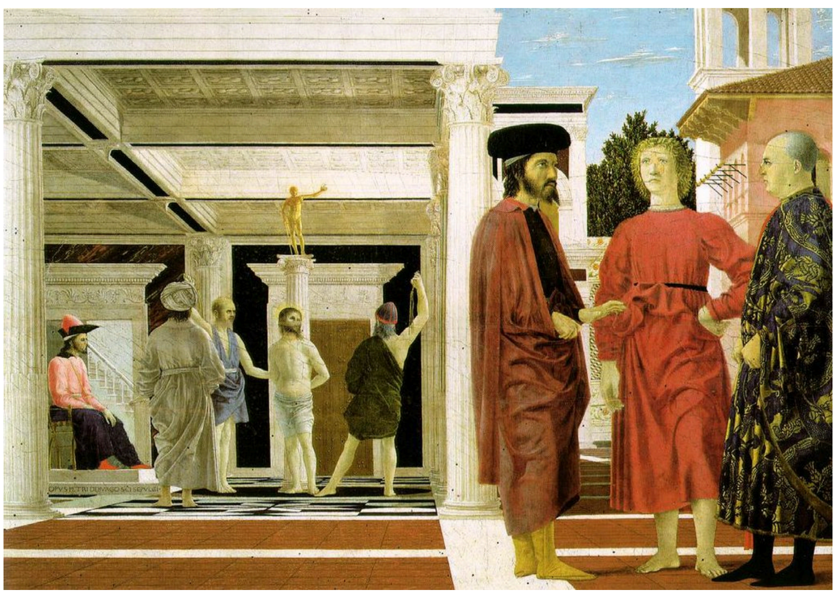

Figure 97: *The Flagellation* by Piero della Francesco (1468-1470). Link to Wikipedia.

Updated: February 19, 2026

Leonardo's *Last Supper* depicts another imaginary room beyond a painted wall. The fresco is painted in the refectory of the Convent of Santa Maria delle Grazie in Milan. At 15 x 25 ft, the *Last Supper* covers the entire end wall of this dining room in the monastery. Leonardo imagines the room at the moment when Christ announces that one of the company will betray him. A dramatic scene ensues. Some disciples rush to protest. Others look imploringly to Christ. Peter pushes toward John to whisper in his ear, pushing Judas forward. Unlike the others, Judas does not look surprised, but suspicious. Leonardo gave a unique human reaction, posture, and motion to each apostle. Christ sits calmly in the middle of the storm.

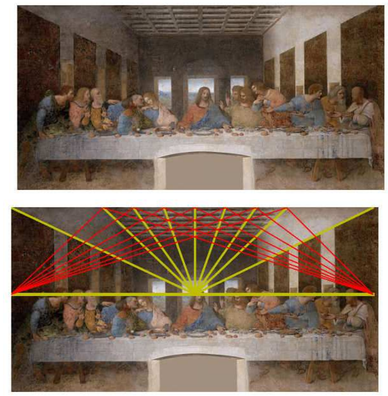

Figure 98: Analysis of *The Last Supper*. The vanishing point at Christ's head can be inferred from the coffered ceiling. Red lines indicate the focus of diagonals two coffers deep. If the coffers are rectangular, twice as wide as long, these foci convey the distance between viewer and image. Kemp (1990).

The architecture of the room allows analysis of its linear perspective. Parallel lines on the coffered ceiling converge to the central point at Christ's head. In the actual painting, there is a small nail hole at the central vanishing point. Leonardo probably pulled a taut string from a nail to make the perfectly radiating lines that emanate outward. But the *Last Supper* is painted above the door of the refectory, much too high for any viewer to put their eye at the central line of sight. Leonardo did not intend an optical illusion, where the Last Supper occurs in a room that extends beyond the refectory wall.

Geometry reveals other ambiguities. The table is slanted downward, improving its view for the audience but making it difficult to eat from. If the ceiling coffers are square (as one might expect), then the room is much deeper than wide. Whether the coffers are rectangular or square, they allow one to calculate the proportional sizes of the wall tapestries. If the tapestries are the same size, the geometry of linear perspective predicts their relative sizes. But the painted tapestries violate Brunelleschi's laws. The tapestry widths diminish in size in the ratios $1{:}\frac{1}{2}{:}\frac{1}{3}{:}\frac{1}{4}$. Scaled to whole numbers (12 : 6 : 4 : 3), these proportions resemble simple harmonic ratios. In the margins of Leonardo's notebooks for the *The Last Supper*, Martin Kemp found a doodled arithmetical progression of tonal harmonies (Fig. 99) [1]. Leonardo traded geometrical harmony for tonal harmony, making little difference to the casual viewer but having some aesthetic meaning to him.

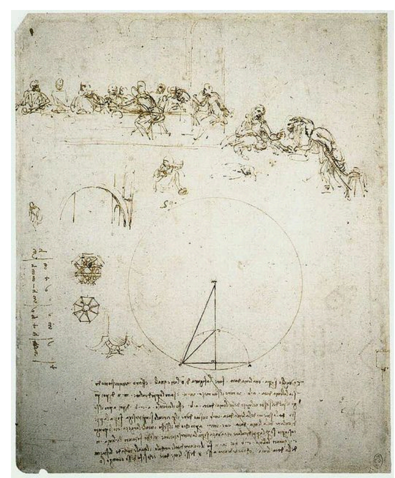

Figure 99: *Study for The Last Supper, with Method of Constructing an Octagon and Arithmetical Calculation*. Leonardo da Vinci.

[1] Martin Kemp. *Leonardo da Vinci, The Marvellous Works of Nature and Man.* Oxford University Press, 2006b

Leonardo favored oil painting which allowed him to work slowly, whereas a fresco (like Masaccio's *Holy Trinity*) is usually made by mixing pigment directly into wet plaster. Traditional fresco painting is long-lasting, but has to be done quickly. So Leonardo made his own medium by mixing tempera and oil. Leonardo's medium proved unstable and the *Last Supper* has sadly decayed

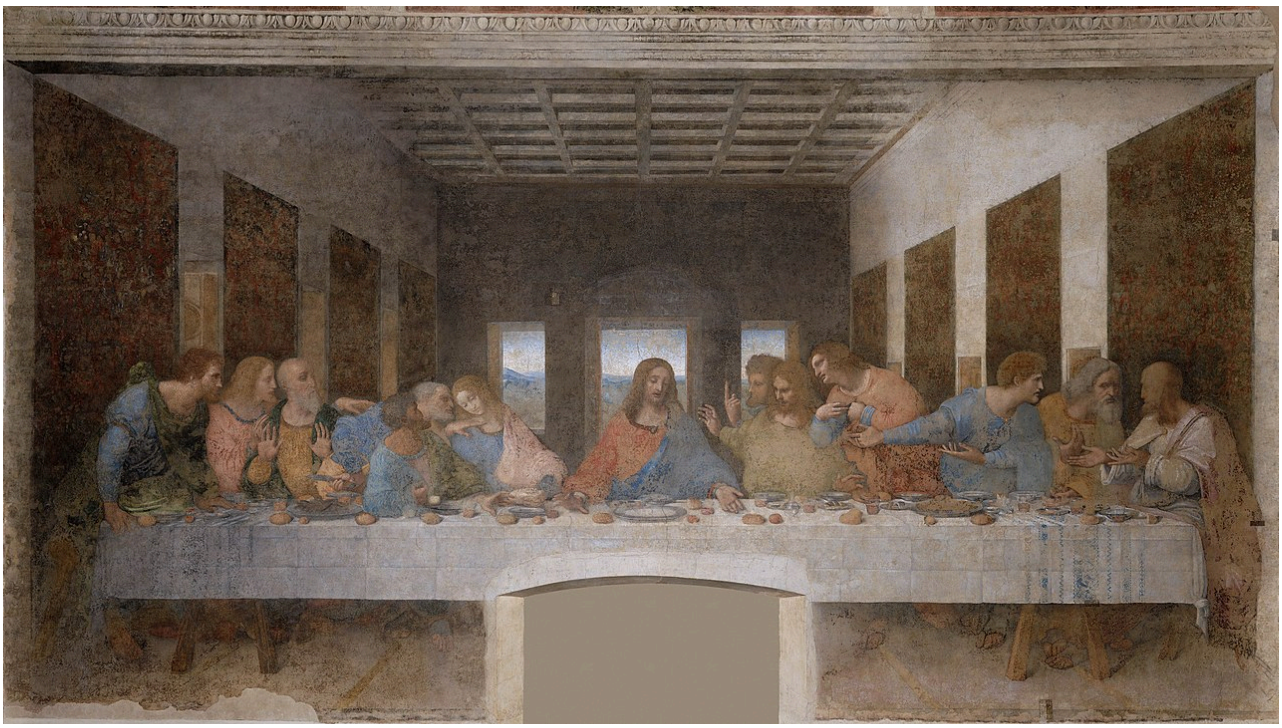

Figure 100: *Last Supper* by Leonardo da Vinci (1495-1498). Link to official site.

GIOVANNI ANTONIO CANAL (1697-1768), known as Canaletto, used linear perspective to make photographically accurate views of Venice, souvenirs for British tourists. The view of Venice in the Harvard Art Museums depicts its principal square, Piazza San Marco, and the domes of the The Basilica of Saint Mark. The sense of linear perspective is strongly enforced by the lines of architecture and patterned pavement.

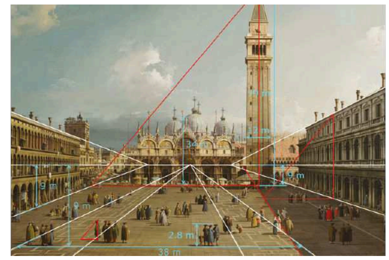

Figure 101: Analysis of Harvard's Canaletto by Erkelens (2019). White lines indicate perspective directions for the Procuratie Vecchie (left), geometric patterns (middle) and Procuratie Nuove (right). Red line and triangles are for analysis of shadows. Computed eye level, heights of the Campanile, Basilica and human figures, and outer width of the patterns are relative to heights of floors of buildings.

GEOMETRIC ANALYSES of Harvard's Canaletto reveal both remarkable fidelity to perspective as well as deviations. The position of the vanishing point predicts the eye level of the painter, about 9 m above the Piazza. Canaletto would have needed to stand on a scaffold. The Basilica is painted too large, and has been artificially magnified. The geometric patterns on the Piazza are faithful to reality. Interestingly, the perspective lines converge to three different vanishing points, all on the horizon. This might be due to the fact that the Piazza San Marco is shaped like a trapezoid.

Figure 102: *Piazza San Marco, Venice* Canaletto, c. 1730-1734, Link to painting

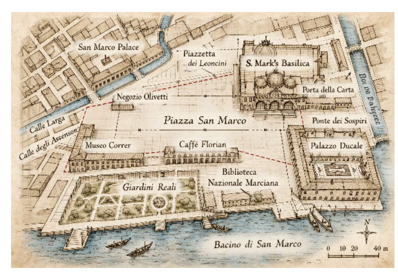

Figure 103: Piazza San Marco is a trapezoid: its flanking buildings diverge as they approach the Basilica. Because the sides are not parallel, they cannot converge toward a single vanishing point. In Canaletto's views of the piazza, this irregular geometry leads to multiple vanishing points, each corresponding to a different directional axis of the square.

Linear perspective is a choice. Chinese painting during the Italian Renaissance had their own tradition of depicting three-dimensional space without linear perspective. The Chinese artist Qiu Ying (1494-1552) rendered architecture according to conventions distinct from single-point projection. Lines that are parallel in the real world are drawn as parallel in the painted world. Instead of visual perspective, this artist chooses an architectural perspective that is arguably more tactile and real. Objects are drawn to look like how they were made.

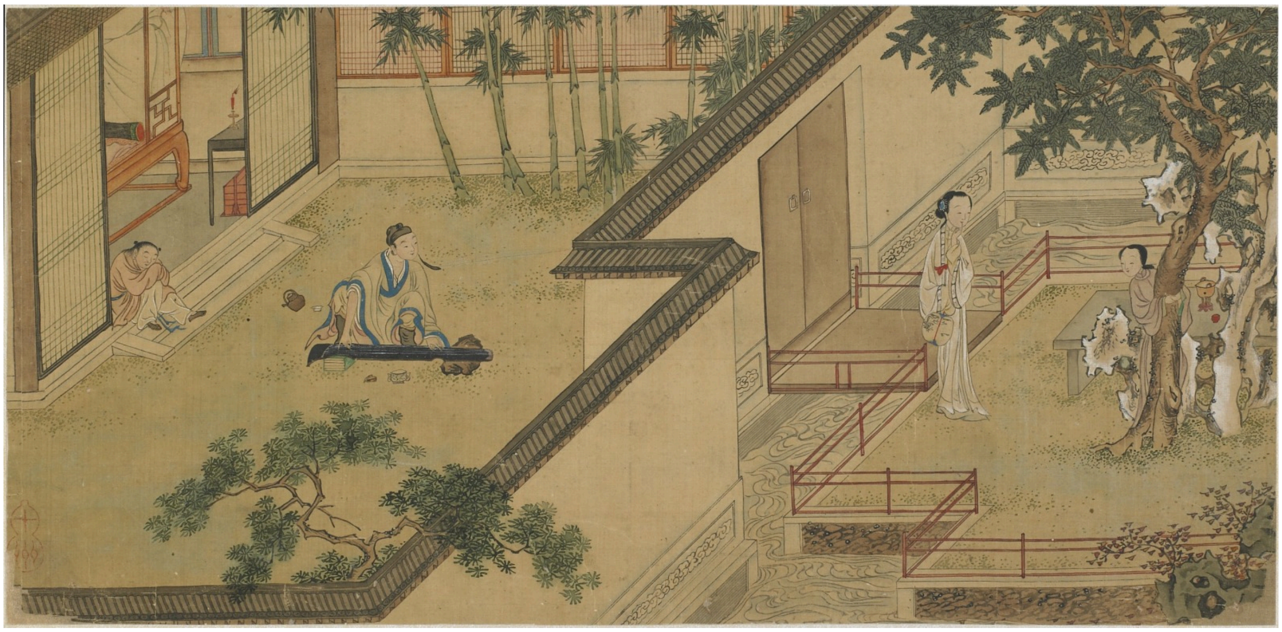

Figure 104: Scenes from *The Story of the Western Wing*, painted by Chinese artist Qiu Ying.

BYZANTIUM thrived for centuries between the Fall of Rome and the Italian Renaissance, preserving much of the scientific and artistic tradition of Greece and Rome. Byzantine artists found their own ways to render space and volume. They often used 'reverse perspective', a technique that captures *more* than linear perspective.

In the *Hospitality of Abraham* (Old Testament Trinity), space does not submit to a single, fixed viewpoint. The artist depicts multiple planes at once. The table is shown as if its top surface, front edge, and side supports can all be seen at the same time. The throne bases tilt forward. The architectural forms open outward rather than receding toward a unified vanishing point. This is an intentional way of conceptualizing the three-dimensionality of an object in a non-optical way.

Florensky has suggested that reverse perspective anticipates the composite viewpoints later explored in Cubism. The perspective is called 'reversed' because depicted objects can be interpreted as being projected from a distant point to the viewing plane, not behind the viewing plane as in linear perspective (Fig. 105).

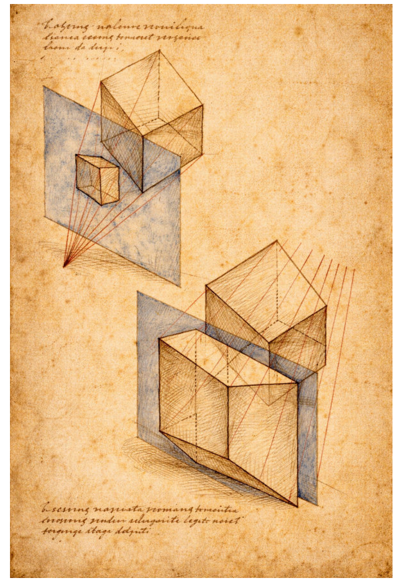

Figure 105: Reverse Perspective Diagram (after Byzantine spatial construction) Study of two cubes intersected by a picture plane, demonstrating the outward divergence of orthogonals characteristic of reverse perspective. Unlike linear perspective, where parallel lines converge toward a vanishing point within the pictorial field, these construction lines expand toward the viewer, implying a center located in front of the image. The upper diagram shows the projection of a smaller cube; the lower, a larger mass whose planes unfold simultaneously.

Figure 106: Andrei Rublev *The Hospitality of Abraham* (Old Testament Trinity), c. 1411 Link to State Tretyakov Gallery, Moscow

David Hockney learned about Byzantine perspective and found ways to incorporate its effects into modern painting. *The Avenue at Middelharnis* by Meindert Hobbema uses trees along an avenue to define strongly converging lines of linear perspective. This landscape is painted within a three-dimensional grid that controls the entire scene. David Hockney made his own version in 2017, *Tall Dutch Trees After Hobbema (Useful Knowledge).* Hockney dissected the scenes onto different panels, turning some panels inside-out by painting them in reverse perspective (Fig. 108)!

Figure 107: *The Avenue at Middelharnis* by Meindert Hobbema, 1689, Link to National Gallery.

Figure 108: *Tall Dutch Trees After Hobbema (Useful Knowledge)* by David Hockney, 2017.

DAVID HOCKNEY quipped that "Photography is all right if you don't mind looking at the world from the point of view of a paralyzed Cyclops." Becoming a paralyzed Cyclops is what happens when the viewer is locked into seeing linear perspective like Brunelleschi. Hockney recognizes that we don't typically look at the world from a single, static viewpoint.

> When a human being is looking at a scene the questions are: What do I see first? What do I see second? What do I see third? A photograph sees it all at once – in one click of the lens from a single point of view – but we don't. And its the fact that it takes us time to see it that makes the space.

Our visual systems not only integrate information from two eyes, but also integrate information over time, dynamically building an internal representation of external scenes as our eyes wander over space. Beyond seeing three-dimensions with geometry, we use additional evidence like shade, luminosity, and context. To 'discover' linear perspective, Brunelleschi had to see like Hockney's paralyzed Cyclops.

Hockney's experiments with reverse-perspective were his conscious efforts to break away from linear perspective. Another experiment was his photocollage of a chair, combining views from different perspectives into one image, retaining the shape of the chair and sense of depth (Figure 109).

Figure 109: *Chair Jardin de Luxembourg* (1985) by David Hockney

References

- Erwin Panofsky. *Perspective as Symbolic Form*. Zone Books, New York, 1991. ISBN 0942299523
- Bevil R. Conway and Margaret S. Livingstone. Perspectives on science and art. *Current Opinion in Neurobiology*, 17(4):476–482, 2007. DOI: 10.1016/j.conb.2007.07.010
- Martin Kemp. *Leonardo da Vinci: The Marvellous Works of Nature and Man*. Oxford University Press, Oxford, 2006a
- Martin Kemp. *The Science of Art: Optical Themes in Western Art from Brunelleschi to Seurat*. Yale University Press, New Haven, 1990. ISBN 0300043376
- David Hockney. *Hockney's Eye: The Art and Technology of Depiction*. Paul Holberton Publishing, London, 2022. ISBN 1913645126
- Pavel Florensky. *Beyond Vision: Essays on the Perception of Art*. Reaktion Books, London, 2002. ISBN 9781861896391

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